A SUPER SIMPLE GUIDE

How Does a Computer Learn to Read Numbers?

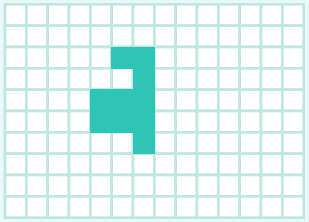
Explaining convolutional layers -- the trick that helps computers spot shapes -- with flashlights, not big words.

Written for a curious reader, by the project's own Claude Sonnet 5 co-author.

CHAPTER 1

Our First Robot Brain

Imagine a picture of a number is really just a big wall of 784 tiny light switches -- some on (bright) and some off (dark). That's all a picture is to a computer: a long list of numbers.



(the teal squares are the 'on' switches)

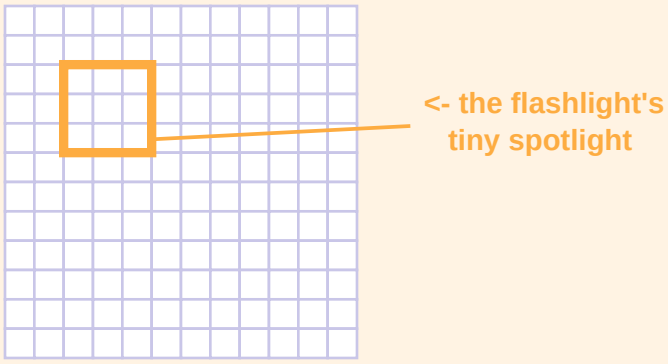
Our first robot brain (the one we built first) works like this: we hire 100 little helpers, and EVERY single helper looks at ALL 784 switches, all at the same time, trying to guess the number.

This actually works pretty well! But it's a little bit silly, if you think about it: every helper has to re-learn, completely on their own, that two switches sitting right next to each other are probably part of the same curve or line. Nobody gets to share that discovery.

CHAPTER 2

The Magic Flashlight

Here's a smarter idea. Instead of one helper staring at the WHOLE wall at once, what if we gave a helper a small flashlight -- one that only lights up a tiny 5x5 patch of the wall at a time?



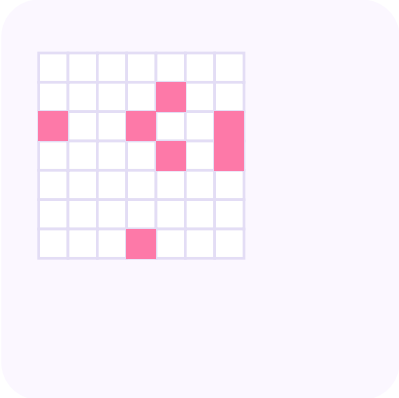
The helper slides this little flashlight across the whole wall, one small step at a time, always asking the exact same question: "does a curvy edge show up right here?"

Because it's carrying the SAME flashlight everywhere -- not a brand new one for every single spot -- it gets really, really good at spotting that one pattern, no matter where it shows up on the wall.

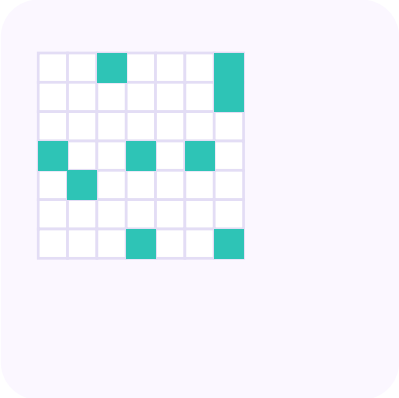
CHAPTER 3

Lots of Flashlights, Lots of Patterns

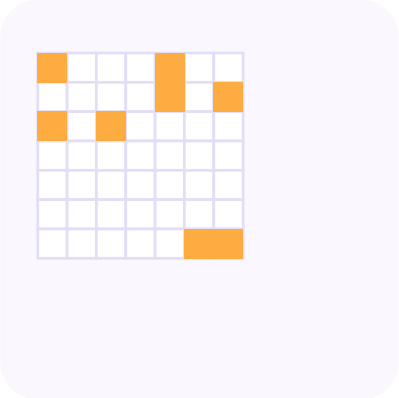
One flashlight finds one kind of pattern. So we hand out a whole box of different flashlights -- maybe 20 of them! One looks for vertical lines. One looks for curvy edges. One looks for corners. And so on.



Vertical-line flashlight



Curve flashlight



Corner flashlight

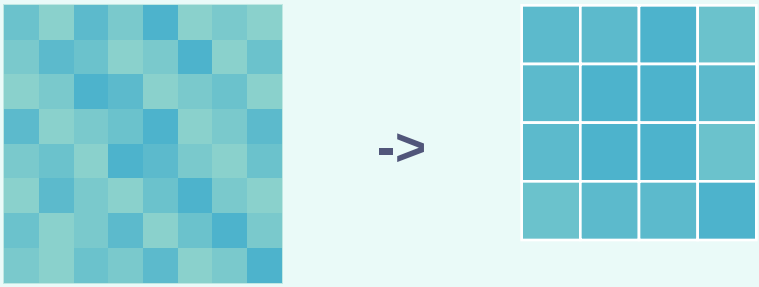
Each flashlight scans the ENTIRE picture and draws its own little map showing exactly where it found its favorite pattern. Bright spots on the map mean "found it right here!"

Grown-up word: each flashlight is called a filter, and the map it draws is called a feature map. A layer that does this scanning is a convolutional layer.

CHAPTER 4

Zooming Out a Little

Now we have a bunch of detailed pattern maps. They're useful, but they're also very precise -- almost TOO precise. If a curve shows up one tiny pixel to the left in a new photo, we don't want the computer to panic and think it's a totally different picture.



So we zoom out: we chop the map into little 2x2 zones, and from each zone we keep only the single brightest square. Everything else gets thrown away.

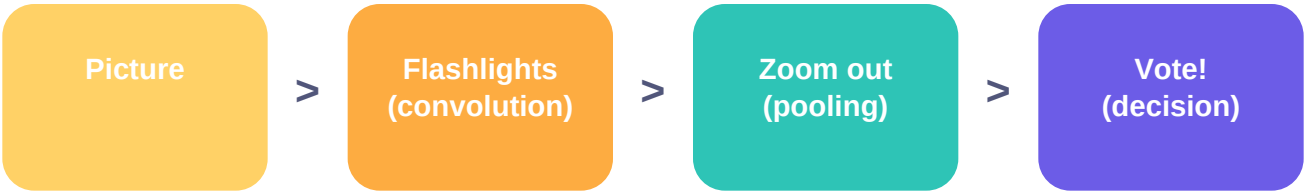
This is called max pooling. It shrinks the map (so there's less to carry around) and makes the whole thing less fussy about tiny shifts in position.

CHAPTER 5

Putting It All Together

After scanning with all our flashlights and zooming out, we're left with a handful of small, simplified maps -- each one a summary of where a certain pattern showed up.

Now we hand these summaries over to the same kind of decision-making team we had before: a group of helpers who look at everything they're given and vote on which digit it probably is.



That's the whole idea! A few rounds of "shine flashlights, then zoom out" followed by one final vote.

CHAPTER 6

Why Is This Better?

1

Fewer things to memorize

Every flashlight is reused everywhere on the picture, instead of every helper needing their own separate rulebook for every single switch. That means far fewer things for the computer to learn.

2

It understands neighbors matter

A flashlight only ever looks at a small, connected patch -- so from the very start, the computer knows that switches sitting next to each other are probably part of the same shape.

3

It finds patterns anywhere

Once a flashlight learns to spot a curve in one corner, it already knows how to spot that exact same curve anywhere else on the page, without needing to be taught all over again.

THE RECAP

So, What's a Convolutional Layer?

It's a layer in a computer's "brain" that scans a picture with a handful of small, reusable flashlights, each hunting for one simple pattern (an edge, a curve, a corner) wherever it appears.

It usually teams up with a "zoom out" step (pooling) to shrink and simplify what it found, before handing everything off to a final voting team that decides what digit it's looking at.

Grown-up footnote: in this project, the plain network (no flashlights, every helper sees every pixel) is called a fully-connected network. The flashlight version is a convolutional neural network, or CNN. Both are built from scratch with NumPy in the mnist-neural-network repo -- read src/network.py for the first kind, and src/conv_layers.py / src/conv_network.py for this one.